

9

NUMBER AND ALGEBRA

SIMULTANEOUS EQUATIONS

Many scientific, natural, economic and social phenomena can be modelled by equations. Often these models consist of more than one equation. For example, when manufacturing milk, equations can be written that describe relationships between quantity, cost and income. These equations can then be solved simultaneously to obtain information about pricing and the quantities that need to be produced and sold to make the most profit.



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Chapter outline

		Working mathematically				
9.01	Solving simultaneous equations graphically*	U	F		R	C
9.02	The elimination method*	U	F			
9.03	The substitution method*	U	F			
9.04	Problems involving simultaneous equations*	U	F	PS	R	C

***PATH**

Wordbank

Quiz
Wordbank 9

coefficient The numerical part of an algebraic term. For example, in $3x^2 + 7x - 1$ the coefficient of x is 7

elimination method A method of solving simultaneous equations that involves combining them to eliminate one of the variables

graphical method A method of solving simultaneous equations that involves graphing them on a number plane and identifying the point(s) of intersection

simultaneous equations 2 (or more) equations that must be solved together so that the solution satisfies both equations. For example, $y = 2x + 1$ and $y = 3x$ are simultaneous equations that have a solution of $x = 1$, $y = 3$

substitution method A method of solving simultaneous equations that involves substituting one equation into another equation

U = Understanding | **F** = Fluency | **PS** = Problem solving | **R** = Reasoning | **C** = Communication

Videos (3):

9.02, 9.03 Simultaneous equations 1 • Simultaneous equations 2

9.04 Simultaneous equations problems

***Twig* video (1):**

9.04 The chase

***PhET* interactive (1):**

9.02 Equality explorer: Two variables

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9.01 Testing simultaneous equations • Intersection of lines

9.04 Simultaneous equations problems

Mind map Simultaneous equations

Puzzles (3):

9.03 Simultaneous equations order activity • Simultaneous equations by substitution

Language of maths Simultaneous equations crossword

Technology (1):

9.01 Simultaneous equations solver

Spreadsheet (1):

9.01 Simultaneous equations solver

Presentation (1):

9.04 Simultaneous equations

Nelson MindTap

To access resources above, visit **cengage.com.au/nelsonmindtap**

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In this chapter you will:

- (PATH) solve linear simultaneous equations graphically, including using graphing technology
- (PATH) solve linear simultaneous equations algebraically using the elimination and substitution methods
- (PATH) solve problems using linear simultaneous equations

Quiz
SkillCheck 9

SkillCheck ANSWERS ON P. 655

1 Given the equation $y = 2x + 5$, find y when:

a $x = 0$ **b** $x = 4$ **c** $x = \frac{1}{2}$ **d** $x = -3$

2 Given the equation $y = 4 - 3x$, find y when:

a $x = 5$ **b** $x = 1$ **c** $x = -1$ **d** $x = -\frac{1}{2}$

3 Test whether the point $(-2, 3)$ lies on the line represented by each equation.

a $y = 1 - x$ **b** $x + y = 3$ **c** $2x - y = 7$

d $\frac{1}{2}x + y = 2$ **e** $y = 3x + 7$ **f** $2y = 3x$

4 **a** Show that the point $(2, 5)$ lies on both the lines $y = 2x + 1$ and $x + y = 7$.

b At what point do these 2 lines intersect?

5 Use the y-intercept and the gradient to graph each equation.

a $y = -2x + 3$ **b** $y = \frac{5}{2}x - 2$ **c** $y = -\frac{4}{3}x + 5$

Investigation

When 2 lines meet

1 Copy and complete the table of values for each equation.

a $x + 2y = 5$

x	−2	−1	0	1	2
y					

b $y = x + 4$

x	−2	−1	0	1	2
y					

2 Which coordinates satisfy both equations?

3 On the same set of axes, draw the graphs of $x + 2y = 5$ and $y = x + 4$.

4 **a** Do the lines you drew in question **3** intersect?

b What are the coordinates of the point of intersection?

5 Repeat questions **1** to **4** for these pairs of equations.

a $x - y = 5$
$2x + y = 1$

b $3x + y = 8$
$x + 2y = 1$

6 Copy and complete.

a The coordinates of the p_____ of intersection between 2 lines satisfy both equations.

b The values of x and y that satisfy both equations are the coordinates of the _____.

Solving simultaneous equations graphically 9.01

A linear equation such as $3x + 5 = 11$ is in one variable (x) and has only one solution ($x = 2$). However, a linear equation in **2** variables, such as $x + 3y = 5$, has more than one solution (for example, $x = 2$, $y = 1$, or $x = 5$, $y = 0$, and so on). The equation actually has an **infinite number** of solutions.

We will now look at solving 2 equations simultaneously to see if there is a solution that satisfies **both** equations. These are called **simultaneous equations**, which can be solved **graphically** or **algebraically**.

Solving simultaneous equations graphically

- Linear simultaneous equations can be graphed as lines on the same number plane.
- If 2 lines are drawn, the lines will intersect (unless they are parallel).
- At the point of intersection, the x-coordinate and y-coordinate represent the solution to the simultaneous equations.

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Worksheets
Testing simultaneous equations

Intersection of lines

Technology
Solving simultaneous equations

Spreadsheet
Simultaneous equations solver

PATH

Example 1

On the same set of axes, graph $3x + y = 4$ and $x + y = -2$, then solve the equations simultaneously.

Solution

$3x + y = 4$

x	0	1	2
y	4	1	−2

$x + y = -2$

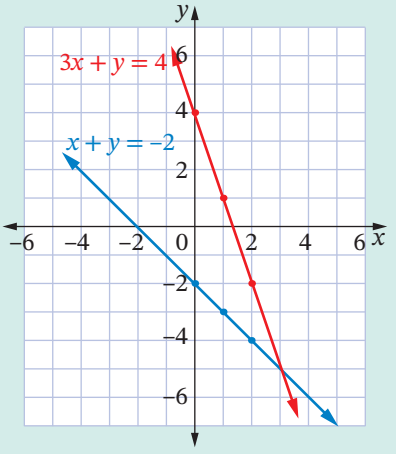
x	0	1	2
y	−2	−3	−4

Graph the equations.

The lines intersect at $(3, -5)$.

$\therefore$ The solution of the simultaneous equations $3x + y = 4$ and $x + y = -2$ is $x = 3, y = -5$.

Check that $x = 3, y = -5$ satisfies both equations.



EXERCISE 9.01 ANSWERS ON P. 655

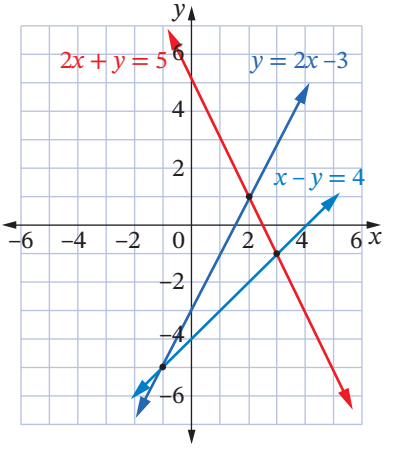
Solving simultaneous equations graphically U F R C

1 Use the graph to write the solution to each pair of simultaneous equations. C

a $x - y = 4$ and $2x + y = 5$

b $2x + y = 5$ and $y = 2x - 3$

c $x - y = 4$ and $y = 2x - 3$



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EXAMPLE 1

9.01

2 Graph each pair of equations together to solve the simultaneous equations.

a $y = 2x$ and $y = 3 - x$

b $y = 2x + 1$ and $y = x - 4$

c $x + y = 3$ and $4x + y = 6$

d $y = -x + 2$ and $y = 3x + 4$

e $y = 2x - 5$ and $y = 5x + 1$

f $2x + y = 6$ and $y = 1 - x$

g $y = 7 - x$ and $y = 3x + 5$

h $x + 2y = 7$ and $2x - y = 4$

i $3x - 2y = 12$ and $x + 2y = 8$

j $y = x + 3$ and $2x - y = 2$

k $5x - y = 5$ and $x + y = 4$

l $5x + 3y = 20$ and $y = x - 4$

3 **a** On the same set of axes, draw the graphs of $y = 1 - 2x$ and $2x + y = 4$.

b Why isn't there a solution to the simultaneous equations $y = 1 - 2x$ and $2x + y = 4$? R C

4 **a** On the same set of axes, draw the graphs of $x + y = 3$ and $2x + 2y = 6$.

b Why are there an infinite number of solutions to the simultaneous equations $x + y = 3$ and $2x + 2y = 6$? R C

Technology

Solving simultaneous equations graphically

You can use graphing technology to solve simultaneous equations graphically.
Write each answer as coordinates in the form (x, y) representing the point of intersection.

1 Use the graphing software to graph these linear equations.

$y = -x + 1$

$y = x + 3$

2 Find the coordinates of the point of intersection of the lines.

3 Repeat steps **1** and **2** to solve each pair of simultaneous equations.

a $y = 2$, $x = 3$

b $y = -2x + 4$, $y = x - 5$

c $y = 5x + 2$, $y = 3x - 1$

d $y = -1$, $x = 0$

e $y = -x - 8$, $y = -3x + 4$

f $y = 2x + 6$, $y = x + 9$

g $2x - y = 5$, $x + y = 4$

h $3x + y = 4$, $x + 2y = 3$

4 **a** Use the graphing software to graph each pair of simultaneous equations.

i $y = x + 4$, $y = x + 6$

ii $y = -2x + 2$, $y = -2x$

b What do you notice about these 2 pairs of equations? Do they intersect?

5 Graph each set of equations and find their point of intersection.

a $y = 3x$, $y = -x + 2$ and $x = 0.5$

b $y = -4x + 1$, $y = -5x$ and $y = x + 6$

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9.02 The elimination method

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Interactive
Equality explorer: Two variables



Video
Simultaneous equations 1

Using graphs to solve simultaneous equations can be time-consuming and inaccurate.

Algebraic methods provide a better way of solving them.
There are 2 algebraic methods: the **elimination method** and the **substitution method**.

In the elimination method, equations are added or subtracted to eliminate one of the variables.

Example 2

Solve the simultaneous equations $x + 3y = 7$ and $4x - 3y = 13$ using the elimination method.

Solution

Label each equation.

$x + 3y = 7$ [1]

$4x - 3y = 13$ [2]

Since there is the same number of ys in each equation, and since they are opposite in sign ($3y$ and $-3y$), add equations [1] and [2] to eliminate the variable y.

$5x = 20$ [1] + [2]

$\frac{5x}{5} = \frac{20}{5}$

$\therefore x = 4$

Substitute $x = 4$ into equation [1] to find the y-value.

$x + 3y = 7$

$4 + 3y = 7$

$3y = 3$

$\therefore y = 1$

$\therefore$ The solution is $x = 4, y = 1$.

Check that this solution works for both equations [1] and [2].

Example 3

Solve the simultaneous equations $2k + 3m = 9$ and $2k - 5m = 1$.

Solution

Label each equation.

$2k + 3m = 9$ [1]

$2k - 5m = 1$ [2]

Since there is the same number of ks in each equation, and because they have the same sign ($2k$ and $2k$), subtract equation [2] from [1] to eliminate k.

$8m = 8$ [1] – [2]

$\therefore m = 1$

Substitute $m = 1$ into equation [1] to find the value of k.

$2k + 3m = 9$

$2k + 3 \times 1 = 9$

$2k + 3 = 9$

$2k = 6$

$\therefore k = 3$

$\therefore$ The solution is $m = 1$, $k = 3$.

Check that this solution works for both equations [1] and [2].

Example 4

Solve $3a + 4c = 8$ and $2a - 3c = 11$.

Solution

$3a + 4c = 8$ [1]

$2a - 3c = 11$ [2]

In this case, neither adding nor subtracting equations [1] and [2] will eliminate a variable. Let's choose to eliminate c. We need to make the **coefficient** of c the same in both equations ($12c$).

The **coefficient** of c is the number in front of the c in the equation

$9a + 12c = 24$ [3] Multiplying both sides of [1] by 3

$8a - 12c = 44$ [4] Multiplying both sides of [2] by 4

$17a = 68$ [3] + [4]

$\therefore a = 4$

Substitute $a = 4$ in [1] to find c.

$3a + 4c = 8$

$3 \times 4 + 4c = 8$

$12 + 4c = 8$

$4c = -4$

$c = -1$

$\therefore$ The solution is $a = 4$, $c = -1$.

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9.02

Video
Simultaneous equations 2

PATH

EXERCISE 9.02 ANSWERS ON P. 655

The elimination method U F

EXAMPLE 2

1 For each pair of simultaneous equations, eliminate one variable by adding the equations, then solve the equations.

a $4k + d = 5$
$2k - d = 7$

b $2x - w = 6$
$x + w = 9$

c $3g + 5h = 4$
$2g - 5h = 6$

d $7p - 4n = -20$
$3p + 4n = 10$

e $4q + 3r = 8$
$-q - 3r = 7$

f $-5k - 3x = 8$
$5k + 4x = -3$

EXAMPLE 3

2 For each pair of simultaneous equations, eliminate one variable by subtracting the equations, then solve the equations.

a $5k + d = 16$
$3k + d = 4$

b $4a + 3c = 7$
$a + 3c = 4$

c $4h + 3y = 24$
$4h - y = 8$

d $3x + 5e = 16$
$3x - 2e = -5$

e $4q - 2w = -1$
$7q - 2w = 8$

f $6p + 5c = 39$
$4p + 5c = 31$

EXAMPLE 4

3 Solve each pair of simultaneous equations.

a $3w + q = 6$
$2w - 3q = 15$

b $2x + m = 5$
$3x + 2m = 3$

c $2d + 3h = 25$
$d + 4h = -5$

d $-3g + 2n = 9$
$g + 5n = 14$

e $5m - h = 10$
$m - 3h = 2$

f $2y + 3e = -6$
$5y - 2e = 23$

g $3q - 2w = 11$
$2q - 5w = 22$

h $5a + 3d = 4$
$4a + 2d = 3$

i $-2p + 3k = 19$
$7p + 4k = 6$

j $5a + 2f = -14$
$2a - 3f = 2$

k $5r - 3c = 2$
$-3r + 2c = -14$

l $5y - 4x = 1$
$2y - 3x = 6$

m $3x + 4y = 20$
$2x - 5y = 21$

n $7g + 3h = 39$
$3g + 5h = 26$

o $5w - 3k = 25$
$3w - 7k = 28$

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The substitution method 9.03

With the **substitution method**, **substitute** the x or y variables from one equation into the other equation.

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9.03

Video
Simultaneous equations 1

Puzzles
Simultaneous equations order activity

Simultaneous equations by substitution

Example 5

Solve the simultaneous equations $y = x + 4$ and $y = 3x - 2$.

Solution

Label each equation.

$y = x + 4$ [1]

$y = 3x - 2$ [2]

Use equation [1] to substitute for y in equation [2] and solve for x.

$$x + 4 = 3x - 2$$
$$-2x + 4 = -2$$
$$-2x = -6$$
$$x = 3$$

Now substitute $x = 3$ into equation [1] to find y.

$y = x + 4$

$y = 3 + 4 = 7$

$\therefore$ The solution is $x = 3$ and $y = 7$.

Example 6

Solve the simultaneous equations $5x + 3y = 9$ and $y = 7 - 3x$.

Video
Simultaneous equations 2

Solution

$5x + 3y = 9$ [1]

$y = 7 - 3x$ [2]

Since y is the subject in [2], substitute equation [2] into equation [1].

$5x + 3(7 - 3x) = 9$

$5x + 21 - 9x = 9$

$-4x = -12$

$x = 3$

Now substitute $x = 3$ into equation [2] to find y.

$y = 7 - 3 \times 3 = -2$

$\therefore$ The solution is $x = 3$ and $y = -2$.

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EXERCISE 9.03 ANSWERS ON P. 656

The substitution method U F

EXAMPLE 5

1 Use the substitution method to solve each pair of simultaneous equations.

a $y = 2x + 1$ and $y = x + 3$

b $y = 5 - 2x$ and $y = 3x + 2$

c $x = 3 + 2y$ and $x = 9 - y$

d $y = -x$ and $y = 3x - 8$

EXAMPLE 6

2 Solve each pair of simultaneous equations.

a $y = 2x + 3$ and $3x - y = 6$

b $y = x - 2$ and $3x + y = 18$

c $y = 1 - 4x$ and $4x + 2y = 3$

d $x = 2y - 5$ and $4x - y = -13$

e $x = 3y - 4$ and $5x - 4y = 2$

f $x = 5 - 3y$ and $4y - x = 23$

g $2x - 5y = -1$ and $y = 10 - x$

h $6y - 2x = 9$ and $y = \frac{x+2}{2}$

i $x = \frac{9-y}{3}$ and $3x + 2y = 10$

j $y = 3x + 5$ and $4x - 3y = 1$

k $3y + 5d - 20 = 0$ and $d = 7 - 2y$

l $w = \frac{3a+2}{5}$ and $2w + 7a = 16$

Investigation



Zero and infinite solutions

1 Copy and complete the table of values for each equation.

a $x + y = 0$

x	-2	-1	0	1	2
y					

b $y = 4 - x$

x	-2	-1	0	1	2
y					

2 Graph $x + y = 0$ and $y = 4 - x$ on the same number plane and find the point of intersection of both lines. What do you notice?

3 Solve algebraically the simultaneous equations $x + y = 0$ and $y = x + 4$. What do you notice?

4 Why is there no solution and why don't the 2 lines don't intersect?

5 Copy and complete this statement:

Parallel lines do not _____ because they have the same _____.

6 Copy and complete the table of values for each equation.

a $2x - 2y = 0$

x	-2	-1	0	1	2
y					

b $y = x$

x	-2	-1	0	1	2
y					

7 Graph $2x - 2y = 0$ and $y = x$ on the same number plane and find the point of intersection of both lines. What do you notice?

8 Solve algebraically the simultaneous equations $2x - 2y = 0$ and $y = x$. What do you notice?

9 Why is there an infinite number of solutions and why are the 2 lines the same?

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10 Copy and complete this statement:

Two linear equations have the _____ graph (called **coincident lines**) if one equation is a _____ of the other equation.

Simultaneous equations that are **parallel lines** when graphed have **no solutions** (and no points of intersection) because the equations are **contradictions** of each other.

For example, $x + y = 0$ can be written as $y = -x$, so the simultaneous equations in Question **3** above can be written as:

$y = -x$

$y = 4 - x$

This means $-x = 4 - x$

or $0 = 4$ which is a **contradiction** and has no solution.

Other examples of contradictions are $3x + 4 = 3x - 1$ and $\frac{2}{x} = 0$.

Simultaneous equations that are the **same line** (coincident lines) when graphed have **infinite solutions** (and infinite points of intersection) because the equations are **identical**.

For example, $2x - 2y = 0$ can be written as $y = x$, so the simultaneous equations in Question **8** above can be written as:

$y = x$

$y = x$

This means $x = x$ which is an **identity** and is true for all values of x.

Other examples of identities are $2x - 8 = 2(x - 4)$ and $x + x = \frac{10x}{5}$.

Investigation

Elimination or substitution method?

With 2 algebraic methods of solving simultaneous equations, often it is more efficient to use one method than the other.

1 Consider these pairs of simultaneous equations.

a $x - 2y = 9$ [1]
$3x + 2y = 11$ [2]

b $4a + 3c = 18$ [1]
$4a - 3c = -6$ [2]

c $3a - 2y = -5$ [1]
$2a + 5y = 3$ [2]

i Why might the **elimination method** be the more appropriate method to use with these equations?

ii What feature in the pairs of equations do you look for to decide if the elimination method is the best one to use?

iii Solve the 3 pairs of simultaneous equations using the elimination method.

2 Consider these pairs of simultaneous equations.

a $m = 2p$ [1]
$m + p = 15$ [2]

b $m = 4 - p$ [1]
$4m - 3p = -6$ [2]

c $p = 2m - 5$ [1]
$5m - 3p = 11$ [2]

i Why might the **substitution method** be the more appropriate method to use with these equations?

ii What feature in the pairs of equations do you look for to decide if the substitution method is the best one to use?

iii Solve the 3 pairs of simultaneous equations using the substitution method.

3 Using whichever method is more efficient, solve each pair of simultaneous equations.

a $7c + 2y = 13$ [1]
$3c + 2y = 1$ [2]

b $m = 5 - k$ [1]
$2m - k = 4$ [2]

c $3x + 8y = 10$ [1]
$x = 3 - 2y$ [2]

d $4h - 3w = 8$ [1]
$4h + 7w = 12$ [2]

e $3d = q$ [1]
$q + 4d = 14$ [2]

f $3h + 5r = 7$ [1]
$2h - 3r = -8$ [2]

9.04 Problems involving simultaneous equations

PATH

Video
The chase

Worksheet
Simultaneous equations problems

Presentation
Simultaneous equations

Sometimes, worded problems can be solved using simultaneous equations. In these situations, following these steps:

- Read the problem carefully
- Identify the variables to be used
- Use the variables to write simultaneous equations from the information given in the problem
- Solve the equations
- Solve the problem by answering in words

Example 7

At an art show there were 520 guests. If there were 46 more women than men, how many women attended the show?

Solution

Let the number of women attending be w.

Let the number of men attending be m.

$w + m = 520$ [1] 520 people altogether.

$w = m + 46$ [2] 46 more women than men.

Use equation [2] to substitute for w in equation [1].

$m + 46 + m = 520$

$2m + 46 = 520$

$2m = 474$

$m = 237$

Substitute $m = 237$ into equation [2] to find w.

$w = 237 + 46 = 283$

$\therefore$ There were 283 women who attended the art show.



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9.04

Example 8

Masi and Amir spent $1582 on shrubs and trees to plant on their large block of land. Altogether they bought 73 plants. The shrubs cost $19 each while the trees cost $32 each. How many of each plant did they buy?

Video
Simultaneous equations problems

Solution

Let x be the number of shrubs.

Let y be the number of trees.

$\therefore x + y = 73$ [1]

and $19x + 32y = 1582$ [2]

Let's choose to eliminate x.

We will need to make

$19x + 19y = 1387$ [3] Multiplying both sides of equation [1] by 19.

$13y = 195$ [2] – [3]

$y = 15$

Substitute $y = 15$ in [1] to find the value of x.

$x + 15 = 73$

$x = 58$

So Masi and Amir bought 58 shrubs and 15 trees.

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EXERCISE 9.04 ANSWERS ON P. 656

Problems involving simultaneous equations U F PS R C

EXAMPLE 7

1 At a school concert there were 640 guests. There were 70 more women than men. How many of the audience were men? R C

2 At a circus, there were twice as many children as there were adults in attendance. Altogether, 1020 attended the circus. How many were children? R C

EXAMPLE 8

3 Tickets to a concert cost \$15 for children and \$28 for adults. Altogether, 650 people attended the concert and ticket sales totalled \$13 052. Let a stand for the number of adults and c stand for the number of children in the audience. R C

a Explain why the equations $a + c = 650$ and $28a + 15c = 13\,052$ correctly match the information.

b Solve the equations simultaneously to find the number of children that attended the concert.

4 Ashleigh bought a total of 130 meat pies and sausage rolls for the canteen of the local football club. Each meat pie cost her \$3 and the sausage rolls were \$2 each. Altogether, Ashleigh spent \$335. How many sausage rolls did she buy? PS R C

5 Tayyab is 3 times as old as Sejuti. The sum of their ages is 48. How old are Tayyab and Sejuti? PS R C

6 The sum of the ages of Hayley and her mother is 70. The difference between their ages is 38 years. How old is Hayley? PS R C

7 A business bought a total of 60 ink cartridges. Some of them were black, costing \$35 each. The others were colour, each costing \$49. How many of each type did the business buy if the total cost of the ink cartridges was \$2422? PS R C

8 The cost of going to the movies for 3 adults and 5 children is \$142, while the cost for 5 adults and 8 children is \$231.50. Find the cost for an adult and the cost for a child. PS R C

9 Pete's Pizzas sells Supreme pizzas for \$15.90 each and vegetarian pizzas for \$13.50 each. If 45 pizzas were sold at lunchtime, totalling \$684.30, how many of each pizza were sold? PS R C

10 Ronel bought 4 punnets of strawberries and 7 punnets of blueberries for \$40.30 and Nikita bought 5 punnets of strawberries and 2 punnets of blueberries for \$26.75. What was the cost of each punnet of strawberries and blueberries? PS R C

PATH

9.04

11 A money box contains only 20-cent coins and 50-cent coins. Altogether, there are 853 coins in the money box and they amount to $281. Let x be the number of 20c coins and y be the number of $2 coins. R C

a Explain why the equations $x + y = 853$ and $20x + 50y = 28\,100$ correctly match the information.

b Solve the equations to determine the number of 20-cent and 50-cent coins in the money box.

12 The initial cost for producing bottles of fresh orange juice is $135 plus $1.20 for each bottle. The bottles of juice are sold for $3 each. C is the cost in dollars, R is the total sales in dollars and n is the number of bottles produced and/or sold. PS R C

a Explain why the equations $C = 135 + 1.2n$ and $R = 3n$ correctly match the information.

b Copy and complete the tables of values below for both equations.

$C = 135 + 1.2n$

n	0	50	100
C			

$R = 3n$

n	0	50	100
R			

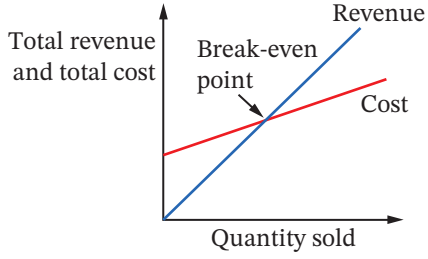
c Draw the graphs of both equations on the same axes for values of 0 to 100 for n on the horizontal axis and values of $0 to $300 on the vertical axis.

d For what value of n is total sales equal to total cost (the break-even point)?

Did you know?

Break-even point

Manufacturers use simultaneous equations to make decisions about how many products they should make and sell. Linear equations can be formed to determine total revenue (the amount made from selling products) and total costs (the cost of making the products).



Total revenue = cost per item × number of items made, while total costs includes rent and production costs.

The equations can be graphed as shown. The point where the 2 lines intersect is called the **break-even point** and occurs when total revenue is equal to total cost.

A publisher receives $35 per book sold. There are fixed costs of $110 000 and production costs per book are $8.50.

a **Determine the equations for total revenue and total costs.**

b **Graph the equations to find the break-even point.**

c **How many books must be sold before the publisher makes a profit?**

Foundation Standard Complex

Quiz
Mental skills 9

Mental skills 9: Maths without calculators ANSWERS ON P. 656

Simplifying fractions and ratios

When simplifying a fraction or a ratio, look for a common factor to divide into both the numerator and the denominator, preferably the highest common factor (HCF).

1 Study each example.

a Simplify 135 : 90.

$135^{27} : 90^{18} = 27 : 18$ Dividing both terms by 5

$27^{3} : 18^{2} = 3 : 2$ Dividing both terms by 9

b Calculate $\frac{3}{8} \times \frac{2}{15}$ in simplest form.

$\frac{3}{8^{4}} \times \frac{2^{1}}{15} = \frac{3}{4} \times \frac{1}{15}$ Dividing 2 and 8 by 2

$\frac{3^{1}}{4} \times \frac{1}{15^{5}} = \frac{1}{20}$ Dividing 3 and 15 by 3

c What fraction is 36 minutes of 1 hour?

$\frac{36}{1 \text{ hour}} = \frac{36 \text{ min}}{60 \text{ min}} = \frac{3}{5}$

2 Now simplify each fraction or ratio.

a $\frac{10}{15}$ **b** $\frac{16}{20}$ **c** $\frac{30}{42}$ **d** $\frac{8}{16}$

e $\frac{20}{80}$ **f** $\frac{6}{36}$ **g** $\frac{20}{24}$ **h** $\frac{12}{30}$

i 20 : 36 **j** 25 : 45 **k** 18 : 40 **l** 28 : 35

m 27 : 21 **n** 16 : 12 **o** $\frac{5}{6} \times \frac{18}{25}$ **p** $\frac{12}{50} \times \frac{10}{21}$

3 Express each as a simplified fraction.

a 425 g of 1 kg **b** 8 months of 1 year **c** 64 cm of 1 m

d 750 mL of 3 L **e** 10 hours of 2 days **f** 80c of $10

Power plus ANSWERS ON P. 656

1 With simultaneous equations in 2 variables, we have 2 equations to solve.
With simultaneous equations in 3 variables, we have 3 equations to solve.

Step 1: Take 2 of the equations and eliminate one of the variables.

Step 2: Take another 2 of the equations and eliminate the same variable.

Step 3: Solve the 2 new simultaneous equations from Steps **1** and **2**.

Step 4: Use substitution to find the values of the other 2 variables.

Use the above steps to solve the following sets of simultaneous equations.

a $2x + y - 3w = -16$
$x - y + 4w = 25$
$3x - y + 2w = 19$

b $3a - 2c + d = 5$
$5a + 2c + d = 25$
$4a + 3c - d = 10$

c $2m + 3n - p = 9$
$3m - 2n + 5p = 27$
$4m + 3n + 2p = 13$

2 **a** Show that the solutions to the simultaneous equations $ax + by = c$ and $dx + ey = f$ are $x = \frac{ce - bf}{ae - bd}$ and $y = \frac{af - cd}{ae - bd}$.

b The above solutions do not work when $ae = bd$. Explain why.

c Solve the equations $3x - 2y = 11$ and $5x + y = 14$ by either the substitution or elimination method. Check that the results in part **a** give the same answer.

d Set up a spreadsheet to solve simultaneous equations of the form $ax + by = c$ and $dx + ey = f$ using the solutions $x = \frac{ce - bf}{ae - bd}$ and $y = \frac{af - cd}{ae - bd}$.

Use your spreadsheet to solve each pair of simultaneous equations.

i $3x + y = 4$
$2x - y = 6$

ii $3x - 5y = 4$
$2x - 3y = 8$

iii $15x + 6y = 17$
$2x + 3y = 8$

CHAPTER 9 REVIEW

Quiz
Language of maths 9

Puzzle
Simultaneous equations crossword

Language of maths

algebraic	axes	coefficient	elimination method
graphical	linear	point of intersection	satisfy
simultaneous equations	solution	substitution method	variable

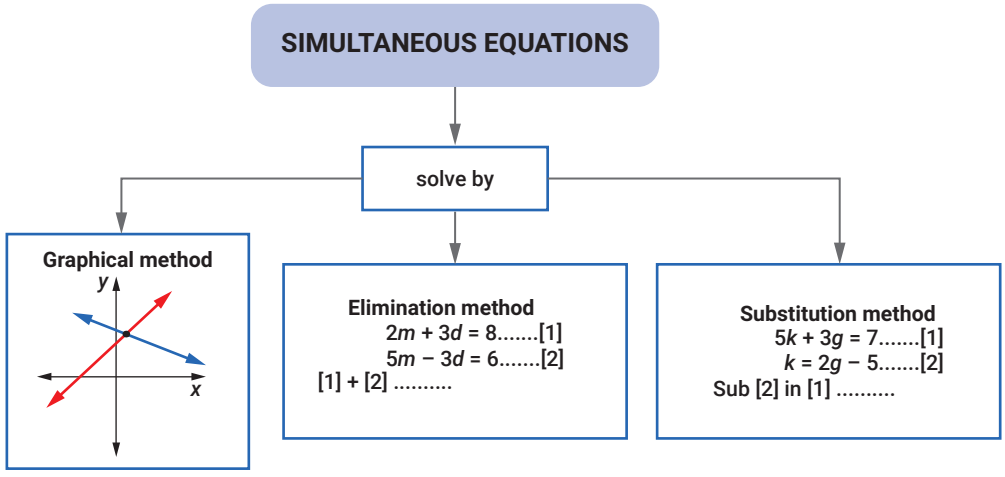
1. How do you think **simultaneous equations** got their name?
2. What are the 2 algebraic methods for solving simultaneous equations?
3. Which algebraic method involves cancelling one of the variables?
4. What word means the answer to an equation or problem?
5. What does '**linear**' mean?
6. Which method of solving simultaneous equations involves finding the point of intersection of lines on a number plane?

Worksheet
Mind map: Simultaneous equations

Topic summary

- In your own words, write down the new things you have learnt about simultaneous equations.
- What parts of this topic did you like?
- What parts of the topic did you find difficult or not understand?
- Copy and complete the following topic overview, and refer to the **Language of maths** word list for keywords you might like to include.

Print (or copy) and complete this mind map of the topic, adding detail to its branches and using pictures, symbols and colour where needed. Ask your teacher to check your work.



TEST YOURSELF 9

ANSWERS ON P. 656

Quiz
Test yourself 9

PATH

9.01

1 Graph each pair of simultaneous equations on the same set of axes.
By finding their point of intersection, write the solution to each pair of equations.

a $y = x + 2$
$y = 6 + 2x$

b $y = 3 - \frac{x}{2}$
$y = 2x - 7$

c $y = 4 - 3x$
$y = x$

d $y = 2x + 3$
$y = 9 - x$

e $x + y = 7$
$y = 2x + 1$

f $y = 5 - 2x$
$y = -1 - x$

9.02

2 Use the elimination method to solve each pair of simultaneous equations.

a $5m + 2c = 6$
$3m + 2c = -4$

b $2x + 3y = 5$
$5x - 3y = 9$

c $3a + 4d = 7$
$3a + d = 4$

d $4x - y = 9$
$x - y = -9$

e $x - 4y = 3$
$x + 2y = -9$

f $3d - 2w = 11$
$2d - 5w = 44$

9.03

3 Use the substitution method to solve each pair of simultaneous equations.

a $y = 7x - 3$
$y = x + 9$

b $m = 4 - p$
$m = -2 + p$

c $h = 3t - 2$
$h = t + 6$

d $a = 4 - 2c$
$a = 6c$

e $x + 2y = 3$
$y = 2 - x$

f $p = 4 - 2q$
$p = 3q + 24$

9.04

4 Solve each problem using simultaneous equations.

a In an audience of 2500, there were 700 more adults than children.
Find the number of adults and the number of children that were in the audience.

b Rosanna bought 30 movie passes as raffle prizes at a school fete for a total cost of $322.50. Movie passes for children cost $8.25 each and the adult price was $14.50.
How many of each did she buy?

c It costs 2 adults and 5 children $191 to go to a football game, while the cost for 3 adults and 2 children is $160. Find the cost of an adult ticket.

d At the cake stall, the student council sell 2 types of cakes: cheesecakes for $4 each and mudcakes for $3 each. In total, they sold 75 cakes for a total of $253.
How many of each cake did they sell?

e In Year 10, there are 213 students. There are 27 more boys than girls.
Find the number of boys and girls in Year 10.